Caruca v2

LLM-based specification mining for opaque shell commands

Naive agent run 9: sort

Claude Code session, PROMPT_v1, claude-fable-5-1, 2026-10-20

SOURCE	/Users/tirandagan/dev/stevens/caruca_v2/naive_llm_caruca/runs/2026-10-20_160340_naive-run-sort_sort_PROMPT_v1/analysis/analysis.md
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PAGES	19

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Run folder: `naive_llm_caruca/runs/2026-10-20_160340_naive-run-sort_sort_PROMPT_v1/`
Written: 2026-10-20, from `stream.jsonl`, `stream.actions.log`, the session transcript, `run.env`, `environment.txt` and the four output files. Call numbers below are the tool-call numbers printed in `stream.actions.log` (1 to 23); times are local (UTC−4). Same setup as runs 4 to 8 (prompt, model, effort, golden image, which has carried the `binwalk` package since run 6); run 7 (`cut`, analysed beside this one) started 2 hours 11 minutes earlier. Runs 6 and 8 (`binwalk`, `convert`) were made and analysed in another session and are not compared here. `sort` **is not one of the four samples.** Checks marked "measured here" were run during this analysis, not by the session: v1's real syntax module and generator on the Mac, and GNU coreutils 9.4 and PaSh's real annotation code (`pash_annotations` 0.2.4) in the `pash` virtual machine.

Summary

A fresh Claude Code session (version 2.1.296, model `claude-fable-5-1` at medium effort, permissions bypassed, web tools and MCP servers removed) in a throwaway Ubuntu 24.04 virtual machine was given prompt version 1 for `sort`. In 304 seconds, 24 turns and 23 tool calls (12 shell, 11 file reads) it produced all four files at a list-price cost of $2.46, with no network use. This is the most thorough of runs 4, 5, 7 and 9. It ran `sort` 88 times before writing anything. After writing, it built two test scripts that ran `sort` about 960 more times: every option spelling in its syntax file, then 20 option sets each split into 2, 3 and 5 parts, sorted part by part, merged with the command its class generates, and compared with one sort of the whole. All of that passed. Its parser table matches PaSh's own `sort` table, argument types included, plus two extra entries (`-y` and `--check`). The session never saw PaSh's table. **But its parallelizability class does not work inside PaSh.** The class offers "deal lines out in turn (round robin), then merge with `sort -m`". PaSh's code permits round robin only when the parts are simply concatenated or joined at their edges. So it raises an error, which PaSh catches silently, and PaSh ends up with no parallelism information for any `sort` invocation except those using `-s` or `-u` (measured here). The session's own tests ran against stand-in modules it wrote itself, which do not have that rule. The format document we gave it does not state the rule either.

What this run teaches

In plain terms, the most useful things run 9 shows. One run, so these are observations to test, not conclusions.

- **Given a harder command, the agent worked much harder, unprompted.** Three times run 7's time and four times its runs before writing. It found real rules of `sort` that are easy to miss: `-o` may appear only once; check mode refuses `-o` and a second file; a bare `--check` takes no value; `-u` also implies "stable"; `--batch-size` must be at least 2. It also knew that sorted pieces must be merged, not just joined, and tested exactly that.
- **Its tests passed, and the class still fails where it matters.** Inside PaSh, its class gives no answer for an ordinary `sort` (no `-s` or `-u`). It could not see this, because it tested against stand-ins of its own making, which only repeat its own assumptions. Run 3, the one run that downloaded PaSh's real package, is the only run whose class tests could have caught a mistake like this.
- **Part of the blame is ours.** The rule that broke the class is in PaSh's source but not in our format document. And PaSh's own `grep` example, one of the samples, contains the same pairing of round robin with a custom merge, though PaSh never reaches it because those merges are marked as not yet available. The agent copied a pattern the example showed.
- **It again matched PaSh's hand-written table without seeing it,** down to the access kind of each file argument: `--random-source` is read as configuration, `--files0-from` names the input files, `-o` is the output, and `-T` is a scratch folder. Like `cut` in run 7, this could come from the manual or from the model having seen PaSh's public files in training; the log cannot tell which.
- **Its report again says more than its tests showed.** Its first test script ended "BAD = 8" and the report says every option "worked". Six of the eight were expected: a bare `--check` correctly takes no value, and checking unsorted input correctly fails. Two came from its own test input. None was investigated or mentioned. The report also says nothing about short options. Its syntax file says they take their value after `=`, but the test script only ever ran them with a separate value.
- **It flagged a real defect and kept it.** It warned that listing `--check` as an option makes PaSh's parser take the next word as its value. That is true (measured here: `sort --check a b` leaves only `b` as a file).
- **Against v1's reference for `sort`, it is close, with one shared weakness.** v1's curated grammar, v1's model output and the agent's file list nearly the same options. Fed to v1's generator, all three produce mostly valid command lines. The agent's does worst (53% accepted at two options per line, against 70% and 85%), for the same reason as `cut`: `=` after short options (`sort -t=:`, `sort -S=1k`), which v1's own curated grammar also does for `-S`.

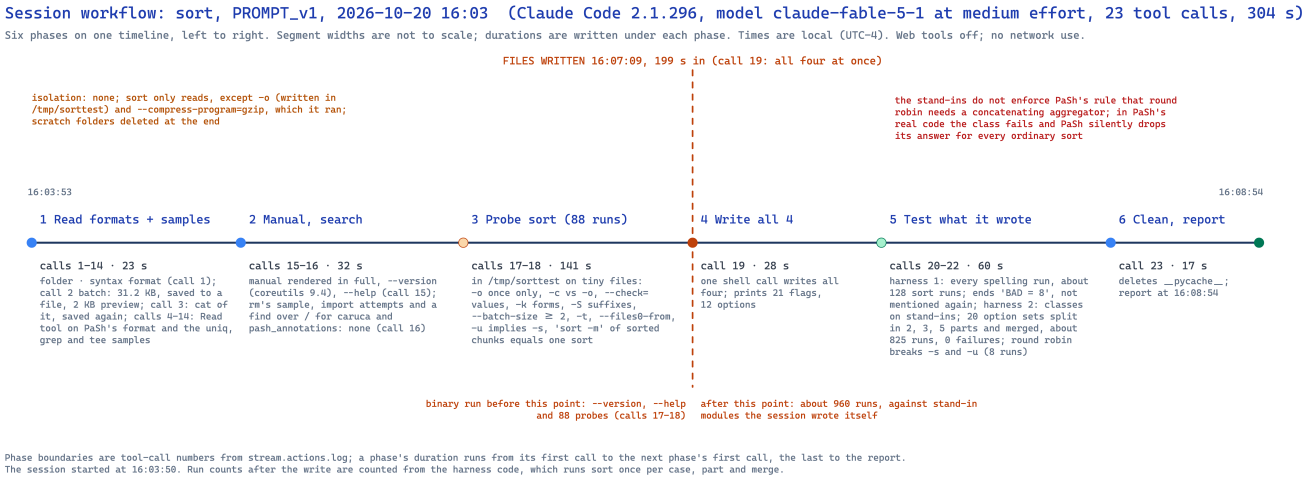
What it means. For a command with real semantics, the agent behaves like a careful engineer who writes tests against a mock of the library instead of the library itself. Its knowledge of `sort` is good and well tested. Its knowledge of PaSh is whatever our documents and examples gave it, and its tests cannot find a gap there. Every one of the agent's runs so far has shown the same thing: what it checks against the binary is mostly right, and what it checks only against its own assumptions is where the defects are.

2.1 Timeline and phases

#	PHASE	CALLS	START	DURATION	WHAT IT READ OR RAN
1	Read formats and samples	1–14 (14)	16:03:53	23 s	the folder and the syntax format (call 1); one batch for PaSh's format and all four samples (call 2), whose 31.2 KB output Claude Code saved to a file, showing a 2 KB preview; `cat` of that file (call 3), saved again; then the Read tool on PaSh's format and the `uniq`, `grep` and `tee` sample files (calls 4–14)
2	Manual and search	15–16 (2)	16:04:16	32 s	the manual rendered in full, `--version` (GNU coreutils 9.4), `--help` (call 15); the `rm` sample, import attempts and a `find` over `/` for `caruca` and `pash_annotations`, none found (call 16)
3	Probe	17–18 (2)	16:04:48	141 s	88 runs of `sort` on small files in `/tmp/sorttest` (61 in call 17, 27 in call 18; counted from the commands)
4	Write	19 (1)	16:07:09	28 s	one shell call wrote all four files and loaded the table
5	Test what it wrote	20–22 (3)	16:07:37	60 s	test script 1 against stand-in modules: every spelling in the syntax file run through `sort` (about 128 runs, counted from the script); ends `BAD = 8` (call 20). Test script 2: both classes on stand-ins; 20 option sets split, sorted, merged and compared (about 825 runs, 0 failures) (call 21); a round-robin check on `-s` and `-u` (8 runs) (call 22)
6	Clean up and report	23 (1)	16:08:37	17 s	removed `__pycache__` in `output/`; report at 16:08:54

First write: call 19 at 16:07:09, 199 seconds into the session (started 16:03:50), after 88 probes. After it, about 960 runs of `sort`, all from its own test scripts. Nothing it wrote was changed after the tests.

Where the time went. API time was 302.6 of the 304.1 seconds. The longest silence was the 52 seconds before the write call: the model composing four files of 196 lines in total.



2.2 Sources it used

Inside the folder. Both format documents; all four samples (the `rm` sample in call 16, the others by the Read tool).

The manual page it was pointed to. Rendered in full in call 15. The table's argument names (`WORD`, `NMERGE`, `PROG`, `KEYDEF`, `SIZE`, `SEP`, `N`) are the manual's.

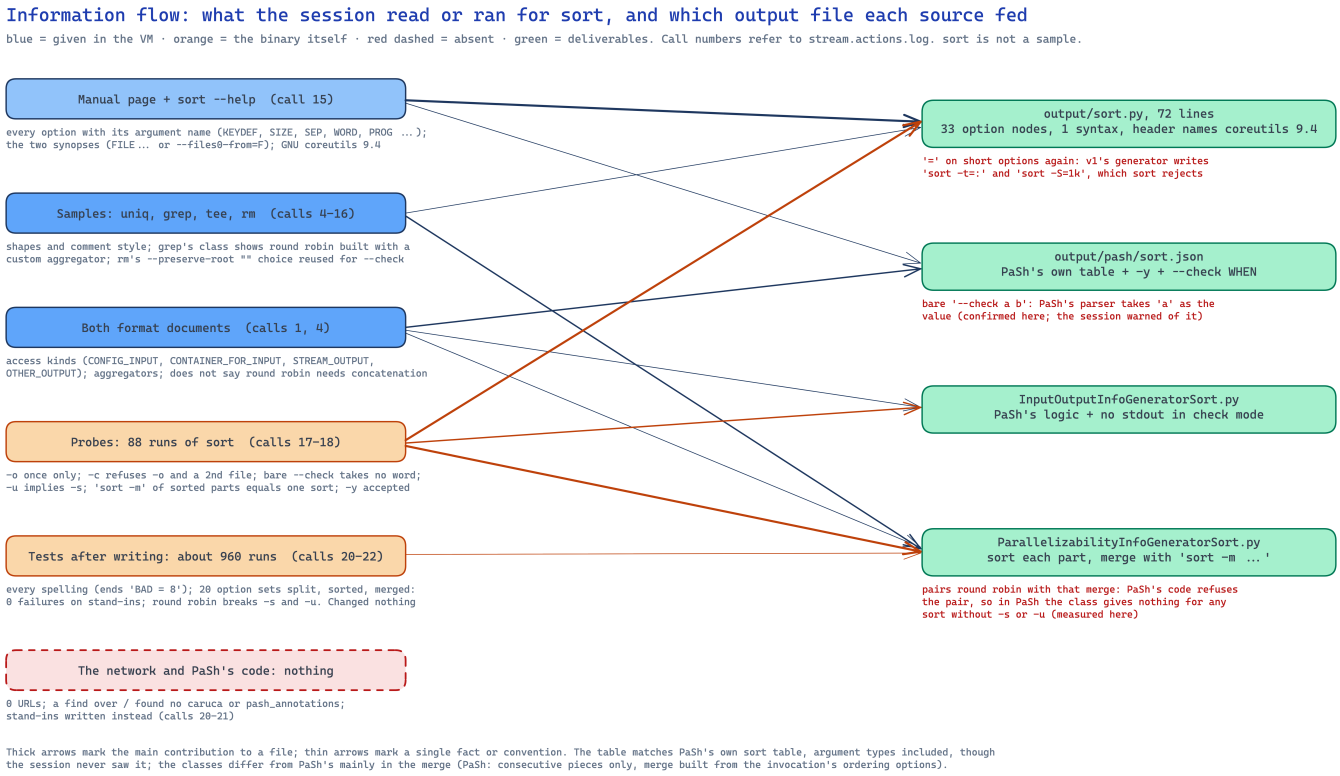
The binary. `--version`, `--help`, 88 probes, then about 960 test runs.

Outside the folder. No network: `bash_urls=0`. A `find` over the whole disk for the two packages (call 16), which found nothing. It did not try to install either.

Compared with PaSh's own `sort` files (from `pash_annotations` 0.2.4; the session never saw them):

FILE	PASH'S OWN	THE AGENT'S
`sort.json`	20 flag entries, 11 options	the same 11 options with **the same argument types** (`["FILE", "CONFIG_INPUT"]` for `--random-source` , `["FILE", "CONTAINER_FOR_INPUT"]` for `--files0-from` , `["FILE", "STREAM_OUTPUT"]` for `-o` , `["DIR", "OTHER_OUTPUT"]` for `-T`), in the same order, plus `--check WHEN` ; the same flags plus `-y` , with `-c` lacking PaSh's `--check` alias
input/output class	standard output unless `-o` ; standard input when there are no operands and no `--files0-from` ; otherwise operands are inputs	the same, plus no standard output in check mode (`-c` , `-C` , `--check`)
parallelizability class	no split for `-c -C -u -z -R -s -m --files0-from --random-source` ; otherwise consecutive pieces only, merged by a 2-way `sort -m` built from the invocation's ordering options; always marked "recombine in any order"	no split for `-c -C --check -m -R --debug -o -z --files0-from` or random sorting; otherwise consecutive pieces **and round robin**, merged by `sort -m` plus the ordering options (written out as a string); with `-s` or `-u` , consecutive pieces only and not "any order"

The agent's class is in one respect more ambitious than PaSh's: it allows `-u` and `-s` (consecutive pieces only), which PaSh's excludes, and it tested that `sort -m -u` over sorted pieces equals `sort -u` of the whole.



2.3 Isolation and safety

`sort` reads, except for `-o`, which it tested on files in `/tmp/sorttest`, and `--compress-program=gzip`, which runs another program. No isolation was set up or needed. It wrote only under `/tmp` and `output/`, and deleted its scratch folders and the Python caches at the end (calls 22 and 23).

2.4 Testing

The 88 probes (calls 17–18) covered multiple files, `-` as standard input, `-o` (once, twice, onto its own input), check mode (`-c`, `-C`, `--check=` with each value, with `-o`, with two files), `-k` repeated and in three spellings, `-T` twice, seven `-S` sizes, `--batch-size`, `--parallel`, every `--sort` word, `-t` (single, double, `\0`), `--files0-from` (three ways, and with an operand), `--random-source`, `--compress-program`, `--debug`, `-m`, `-u`, `-z`, grouped flags, `-y`, `--`, the obsolete `+POS` form, and whether merging sorted pieces reproduces one sort (with `-u` and `-n`).

After writing (calls 20–22), as described in the timeline. Test script 1 checks spellings against the binary but runs short options only with a separate value, never in the `=` form the syntax file states. Test script 2 checks the merge command the class generates against the binary (0 failures over 20 option sets). It checks PaSh's side only against stand-ins: its `make_parallelizer_round_robin` returns a tuple and enforces nothing.

The closing report's claims, checked against the log.

CLAIM	SUPPORTED?
“Every spelling in the spec was run against `/usr/bin/sort`”	Yes (call 20), but short options only in the separate-word form
“all options rejected a missing argument, and both `--long=value` and separate-word forms worked”	**No.** The script ended `BAD = 8`: six `--check` cases (bare `--check` takes no value, which the session knew; checking unsorted test input exits 1) and two `--files0-from` cases (its test list was empty). Not mentioned in the report
“Spec, JSON table and `--help` output agree on the full set of spellings”	Yes (call 20)
“`-o` at most once, `-c` incompatible with `-o` and a second operand, bare `--check` leaves the next word as an operand, `-k` and `-T` repeat, `--batch-size` minimum 2, `--parallel` minimum 1, `-t` rejects multi-character separators”	Yes (calls 17–18)
“20 option combinations were split into 2, 3 and 5 parts … All matched the sequential output, including reversed merge order where commutativity is claimed”	Yes (call 21, `end-to-end failures: 0`), against stand-ins
“A negative test showed round-robin does break `-s` and `-u`”	Yes (call 22)
Doubt: “If the real option objects use different accessors, the merge command would lose its arguments”	**Resolved, no problem** (measured here): PaSh’s options have `get_name()` and `get_arg()`, and inside PaSh the class produced `sort -m -s -k 1,1` for `sort -s -k 1,1`
Doubt: “PaSh’s simple parser would wrongly swallow the next word after a bare `--check`”	**Confirmed** (measured here): `sort --check a b` parses as `--check` with value `a` and the single file `b`
(not in the report) whether the classes run in PaSh	**They do not, for the common case** (next section)

2.5 Reproducibility and versions

- **Recorded during the run:** `sort --version` (GNU coreutils 9.4). The OS was not queried; `run.env` records Ubuntu 24.04.4 LTS, aarch64.
- **In the output:** the syntax file's header names GNU coreutils 9.4 and lists what was verified; the table and classes do not (1 of 4 files, as in runs 5 and 7).
- **Repeatability:** each probe ran once; the merge test was deterministic (fixed random seed for its test data).

2.6 Deliverable quality, mechanically

- All four files parse. `output/sort.py` **imports under v1's real** `caruca.ir.syntax`: 1 syntax, 33 option nodes, 1 positional `Path` (zero or more, `-` is standard input).
- **Against the manual:** the session's own script found no spelling in `--help` missing from the syntax file or table. `-y` is not in the manual; `sort` accepts it.

Inside PaSh's real code (measured here)

A copy of `pash_annotations` 0.2.4 was made in the `pash` VM with the agent's three files laid over PaSh's own, so the installed package was not changed. Twelve `sort` invocations were run through PaSh's parser and its two lookup functions, once with PaSh's own files and once with the agent's.

INVOCATION	PASH'S OWN FILES	THE AGENT'S FILES
`sort a.txt`, `sort -n -k 2 a b`, `sort -t : -k 2,2n -r`	consecutive pieces, merge built from the ordering options	**no answer**: the class raises an error that PaSh catches and discards
`sort --sort=numeric a`, `sort --random-source=x a`, `sort -k2 a`	consecutive pieces (the attached values stay among the files, for both)	**no answer**, same cause
`sort -u`, `sort -s -k 1,1`	no split	consecutive pieces, merged by `sort -m -u` or `sort -m -s -k 1,1`
`sort -o out a`	consecutive pieces	no split
`sort -m a b`, `sort -R a`, `sort -c a`	no split	no split

The input/output class gave an answer in every case. The error is an assertion in PaSh's `Parallelizer`. A round-robin split may be paired only with concatenation or with one of the three "adjacent lines" aggregators. The agent pairs it with a custom two-way merge. PaSh catches every exception from a class and returns "no information", so the failure is silent. The consequence is that PaSh treats the most common `sort` invocations as if no annotation existed. Neither our format document (`formats/pash_annotation.md`, which lists the splitters and aggregators but no rule about combining them) nor the session's stand-ins carried the rule. PaSh's own `grep` sample builds the same combination, inside a branch that never runs because its custom aggregators are marked not implemented.

How v1's generator reads the three syntax files (measured here)

As for `cut` in run 7: each syntax file was fed to v1's own generator, and every distinct command line was run once on GNU coreutils 9.4 with a three-line input. Lines whose printed form a shell cannot re-read, or that use v1's path placeholders or an empty file list, are left out. At the shipped limit of 4 options per line v1's reference grammar alone yields 307,023 distinct lines (1,972,046 in all), so the run was made at 1 and 2.

SYNTAX FILE	LIMIT 1: JUDGED · ACCEPTED · REAL WORK	LIMIT 2: JUDGED · ACCEPTED · REAL WORK	MAIN REJECTIONS AT LIMIT 2
v1's curated reference	53 · 45 (85%) · 43	1,275 · 894 (70%) · 803	`--batch-size` and `--parallel` given –1, 0 or a word; `-S=` values
v1's model output	44 · 42 (95%) · 40	903 · 771 (85%) · 686	`--parallel` –1 or 0; incompatible ordering flags (`-gV` …)
the agent's (run 9)	52 · 39 (75%) · 37	1,200 · 636 (53%) · 551	`-t=:` (a two-character separator), `-S=1k`, `-k=…`, `--check=` (empty)

"Real work" means accepted and not carrying `--help` or `--version`. Unlike `cut`, every file here yields plenty of usable lines, because `sort` needs no option at all. The agent's file loses most where it puts `=` after a short option, and through the empty `--check=` its rm-style "bare spelling" choice produces. v1's reference loses through its value generator (integers –1, 0 and 1 for options that need at least 1 or 2).

- Not done here: scoring the syntax file against a reference.

2.7 Cost and effort

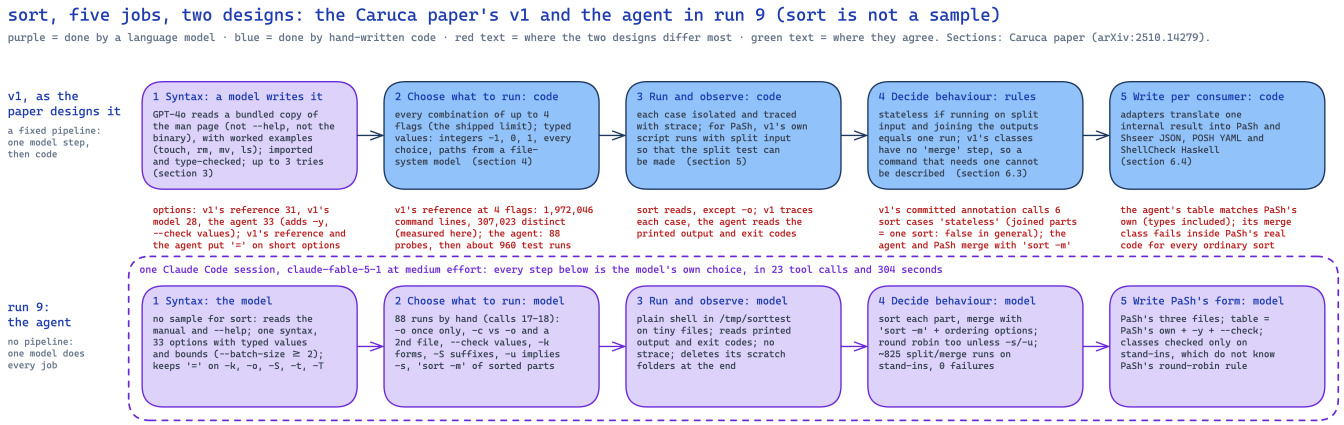
MEASURE	VALUE
Model	`claude-fable-5-1`, medium effort (list-price cost basis, billed to the Max subscription)
Turns	24
Tool calls	23: 12 Bash, 11 Read
Wall clock	304.1 s (API time 302.6 s); 16:03:50 to 16:08:54
Input tokens (uncached)	386
Cache writes	55,629 tokens
Cache reads	571,915 tokens
Output tokens	23,977, of which 6,381 thinking
Reported cost	$2.46 at list price
Permission denials	0 (permissions bypassed)
Web search / fetch tools	not available; 0 URLs in shell commands

Not included: the launcher’s $0.02 trial call that chose the model.

2.8 Behaviour worth recording

- **The output-size limit cost it a turn again** (calls 2 and 3), as in run 7; it then read file by file.
- **It searched the whole disk for the two packages** (call 16), then wrote stand-ins, as in run 7. It did not try to install them; run 3 (a different model) did.
- **It tested one of its own restrictions.** It wrote a test to show that round robin really does break `-s` and `-u` before keeping its class's exclusion of them, rather than only testing the cases it allows.
- **It reused a sample's device in a new place.** It modelled `--check`'s bare spelling the way the `rm` sample models `--preserve-root` (an empty choice) and said so in a comment. Under v1's generator that produces `sort --check=`, which `sort` rejects.
- **No questions asked**, nothing written outside `/tmp` and `output/`.

How the Caruca paper designs the experiment, and what the agent chose instead



Scale for sort. v1's generator, at its shipped limit of 4 flags, writes 1,972,046 command lines from its reference grammar, 307,023 of them distinct (measured here, not run).
The agent ran sort 88 times before writing and about 960 times after, all hand-chosen or built by its own test scripts.

v1 is shown as the paper designs it (sections 2–6), not as run for this comparison. Counts marked 'measured here' come from v1's own generator and GNU coreutils 9.4 in the pash VM. Method, not quality, is compared.

JOB	V1, AS THE PAPER DESIGNS IT	RUN 9'S AGENT
1. Learn the argument syntax	A model reads a bundled man page with worked examples (`touch`, `rm`, `mv`, `ls`) (section 3). v1's model output for `sort` has 28 options; v1's curated reference 31	Manual, `--help` and 88 probes; 33 options with value bounds and the `--check` values
2. Choose what to run	Code enumerates option combinations up to 4 with typed values (section 4): for `sort`, 1,972,046 command lines, 307,023 distinct (measured here)	88 hand-chosen probes, then test scripts (about 960 runs)
3. Run and observe	Each case isolated and traced; split-input runs for PaSh (section 5)	Plain shell on small files; output and exit codes
4. Decide the behaviour	Rules over traces. v1's annotator assigns only `stateless`, `non-pure` or `side-effectful` (`tracer/data.py`): there is no class for "parallel with a merge step"	Reasoned that sorted pieces need a merge, tested `sort -m` against one sort for 20 option sets
5. Write for consumers	v1's committed `sort` annotation (`caruca/save/sort.json`, 2024-12-08): 6 cases (`sort`, `sort -m`, each with no file, a relative and an absolute path), all `stateless`, standard output only	PaSh's current Python form; table matches PaSh's own; parallelizability class fails inside PaSh for ordinary sorts

`sort` is the case where a merge step decides everything. Splitting the input and joining the sorted pieces does not give a sorted whole, so `sort` can be parallelised only with a merge. PaSh's own annotation says exactly that, and so does the agent's. v1's committed annotation calls six `sort` cases `stateless`, which by v1's own definition means "split, run, join gives the same result". That is false for `sort` in general. *Not established:* why v1's traces led there. One possible mechanism, from v1's code: when combining environments, v1 keeps the most parallelizable class after dropping failed runs, so small or already-ordered test inputs could produce it. Not checked against the trace for `sort`.

Where the agent's design is weakest is the consumer, not the command. Every claim it made about `sort` that it checked against `sort` holds. The one that fails is a claim about PaSh that it could only check against its own imitation of PaSh.

Cost. The paper reports CPU time per command (section 7.4), not the cost of its model step. Run 9 took 304 seconds and $2.46 at list price.

Run 9 compared with runs 4, 5 and 7

Same model, effort, prompt, image and day; different command.

MEASURE	RUN 4 (`GREP` , SAMPLE)	RUN 5 (`RM` , SAMPLE)	RUN 7 (`CUT`)	RUN 9 (`SORT`)
Wall clock · turns · tool calls	322 s · 11 · 10	60 s · 6 · 5	104 s · 18 · 17	304 s · 24 · 23
Output tokens (thinking) · cost	24,495 (6,663) · $2.54	4,623 (937) · $0.80	9,117 (1,464) · $1.25	23,977 (6,381) · $2.46
Binary runs before writing	72	27	21	88
Test after writing	154 runs, stand-ins	none	45 runs, stand-ins	about 960 runs, stand-ins
Parser table versus PaSh's own	the sample's	the sample's	PaSh's minus `-M` (unseen)	PaSh's plus `-y` , `--check` (unseen)
Classes inside PaSh's real code	not run	not run	PaSh's answers except `-z`	no answer for ordinary sorts
Version named in output files	3 of 4	1 of 4	1 of 4	1 of 4
Report overstates	"Everything passed" (1 failure)	bare `--interactive` (manual)	"every option … matched"	"worked" over `BAD = 8`

What this run suggests we ask next

1. **State PaSh's combination rule in the format document**, or give the session the real `pash_annotations` package. Either is a repair to our materials, not tuning: the request as written allowed an answer PaSh cannot accept. Record it in task 016's revision log if changed.
2. **Load every run's classes in the pash VM as a standard step** of the analysis. Runs 4 and 5 were never checked this way.
3. **Score syntax files as generator input too** (share of generated lines the binary accepts), as the analyses of runs 6, 7 and 9 did.
4. **Look at v1's `stateless` label for `sort`** in its traces before citing it in any comparison.

Diagram tooling

The three diagrams were drawn with the Excalidraw diagram skill (`https://github.com/coleam00/excalidraw-diagram-skill`, at `.claude/skills/excalidraw-diagram/`). **The skill's repository declares no licence**; it is listed as third-party, unlicensed material in `LICENSE-TEMPLATES.md`, and is kept in the repository by Tiran's decision of 2026-10-13. Elements were placed by hand in the `*.spec.json` files beside each diagram and expanded with `naive_llm_caruca/excalidraw_fill.py`; the v1-versus-agent layout is run 7's with run 9's content. Each was rendered with `uv run python render_excalidraw.py <file>`, inspected, and corrected.